

Manual

Online Visualization of Power Values EGM-OVL 8.1

Version 1.0.23049

Last changed on: 01.09.2020

Copyright

©Copyright 2020 All rights reserved.

SAP® and R/3® are registered trademarks of SAP AG.

WINDOWS® is a registered trademark of Microsoft Corporation.

MPDV® and HYDRA® are registered trademarks of MPDV Mikrolab GmbH.

ORACLE® is a registered trademark of ORACLE Corporation, California, USA.

Copying and distribution of this documentation or any part thereof, for any purpose or in any form, is prohibited without prior written permission from MPDV Mikrolab GmbH.

The information contained in this documentation is subject to change without prior notice.

Contents

1	Online Visualization of Power Values	4
2	Graphic Process Analysis (based on machines)	5
3	Process Trend.....	8
4	Process Data Monitor.....	10
5	Process Visualization	12

1 Online Visualization of Power Values

Purpose

Graphic online visualization of performance values and process parameters collected using the power meters.

You use the function package when:

- You wish to display the currently collected performance values from the power meters online at a terminal or the interfaced HYDRA MPC clients.

Integration

The EMG-LEE basic function is a precondition for the evaluation.

Features

Display of performance values collected at one or more meters:

- Permanent display of the performance values for one or more collection points
- Independent creation of individual system charts, process charts, etc. with the editor for process visualization (MDS-PVE)
- Display of system charts, process charts, etc. created during the course of HYDRA customizing
- Online display of the performance values at the client with predefined pointer graphics
- Online display of the performance values as a trend line; display of the last measured values and the progress as a continuous trend chart

2 Graphic Process Analysis (based on machines)

Summary

Menu	Quality Management → Process Analysis → Graphic Process Analysis (based on machines)
Transaction code	gpdam
Function authorization	gpda

The graphic process analysis is used for the process analysis in quality management.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time period and selection criteria and to compare them with each other. In addition, the measured values can be displayed together with the machine status or binary process events. The selected process value lines and events can also be saved together as a compilation under a freely available name so that analyses that recur often can be called up easily.

Integration

The function displays collected measured values of the process characteristics, events and machine statuses. These values must have been previously collected in the system.

Selection criteria

The following selection criteria are available in the application:

Machine

The required workplace can be searched specifically in the "general" index tab using the pool of workplaces application.

Time domain from – until

Specifies a specific period of time for the evaluation.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Tag type + Tag ID

Selection of the tag type (in a user-friendly manner by a drop-down list) and the tag ID.

Consider long-term data

Long-term data may also be taken into account by clicking this checkbox.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Toolbar



Show process values

Show or hides the analog trace.



Show machine status

Shows or hides the machine status (machine time profile). This is shown in the lower section of the window.



Show events

Shows or hides the bit trace (events). This is shown in the lower section of the window.

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons



can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. Or an area inserted for graphic process analysis is removed.

A special feature of this graphic is the ability to zoom in and out using keyboard and mouse combinations. To zoom in on the graphic, press the SHIFT key and hold it down and then use the left mouse button to highlight an area of the graphic. After the key is released, this area is enlarged in the display.

To zoom out again, press and hold the ALT key down. Now the mouse pointer appears as a magnifying glass with a minus sign. By clicking in the graphic, the view zooms out again and the image appears smaller.

If the machine time profile is shown, the user also can carry out selections for the graphic of the machine time profile there. In the machine time profile, the machine status is always displayed. If resource statuses are used at the same time, they can also be shown.

If the events are shown, individual events can be selected there.

Display configuration

The following configuration options can be set for each chart:

Checkbox “Legend”

A legend may be shown for the data displayed in the diagram by checking this option. In case a legend is shown, axis labeling is automatically hidden to save space.

Checkbox “Show scale”

The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show character designation”:

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!

3 Process Trend

Summary

Menu	Quality management → Process data collection→ Process trend
Transaction code	tvpd
Function authorization	tviewpd

The process trend is used to perform process monitoring in quality management.

Usage

The collection processes defined through the collection rules allow to visualize the defined characteristics online. The characteristics identified for visualization are provided to the network by the related data servers. The visualization components existing in this application connect to these servers and will show the measured values online.

The presentation is made as trend line.

Integration

The display functions will visualize the data of the involved shop floor servers in accordance with the collection specifications . In doing so, always a certain amount of historic measured values will be shown and updated by a current value.

Requirements

It must be possible to reach the shop floor servers involved in the network and they must be configured for the presentation of the process data. A registration service (yellow pages to visualize process data online) must be set-up in the network. The display layouts must be available locally or on the server.

Selection parameters

The following selection criteria are available in the application:

Workplace

The pool application workplace can be used to select the desired workplace.

Process parameter

The process parameter drop-down list offers a selection of process parameters.

Workplace

The pool application workplace can be used to select the desired workplace.

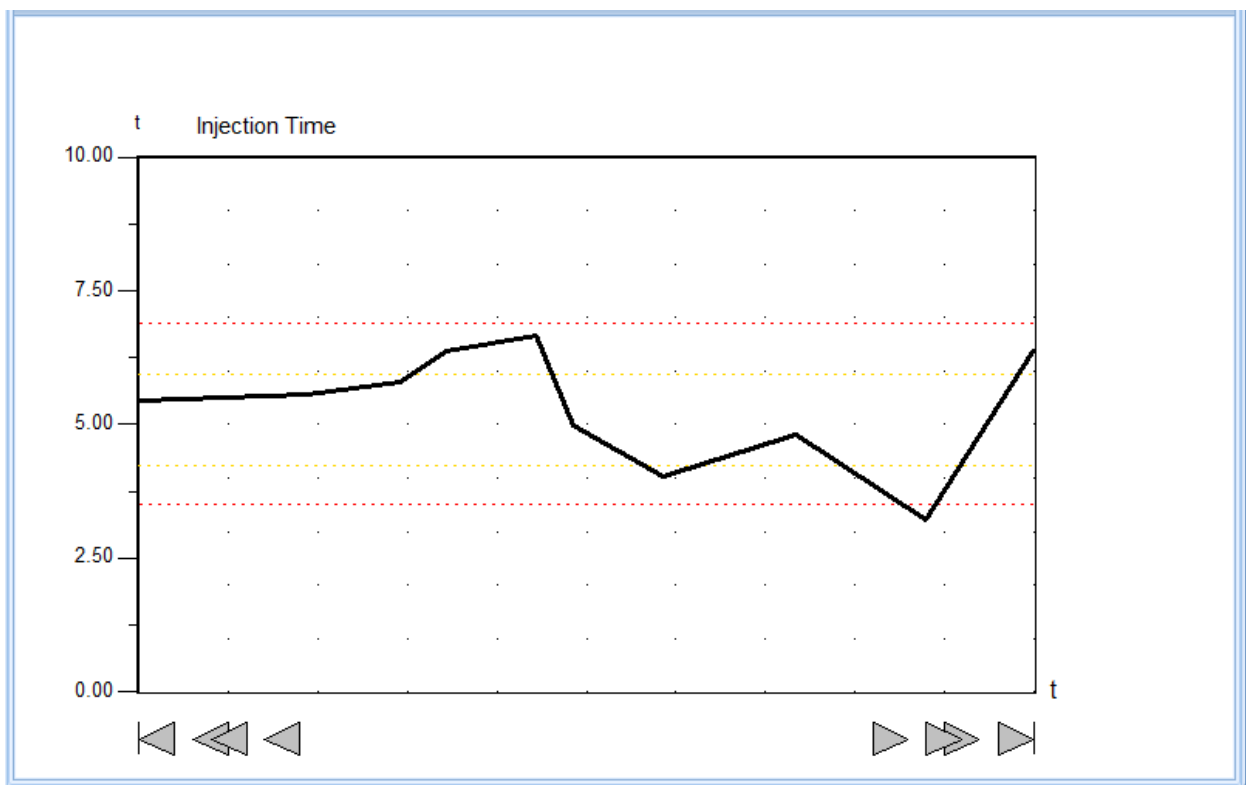
Process parameter

The process parameter drop-down list offers a selection of process parameters.

Field descriptions

Detail applications

The visualization area is used to show the graph of the selected workplace and the individual selected process parameters.



4 Process Data Monitor

Summary

Menu	Quality management → Process monitoring → Process data monitor
Transaction code	mvpd
Function authorization	mviewpd

The process data monitor is used for the process monitoring in quality management.

Usage

The collection procedures defined by the collection rules also allow the online visualization of defined characteristics. The characteristics identified for visualization are provided by the respectively related data servers in the network. The display components existing in this application are connected to these servers and display the measured values online.

The display takes the form of a vector diagram.

Integration

The data of the participating shop floor servers is visualized by the display function based on the specifications of the collection requests. A certain number of measured values from the past are always displayed as well and are part of a continuation including the current value.

Requirement

The participating shop floor servers in the network must be accessible and they must be configured for the presentation of process data. A registration service (yellow pages for process data online visualization) must be set up in the network. The layouts for the display must be available locally or on the server.

Selection criteria

The following selection criteria are available in the application:

Workplace

In the workplace pool application, the desired workplace can be selected.

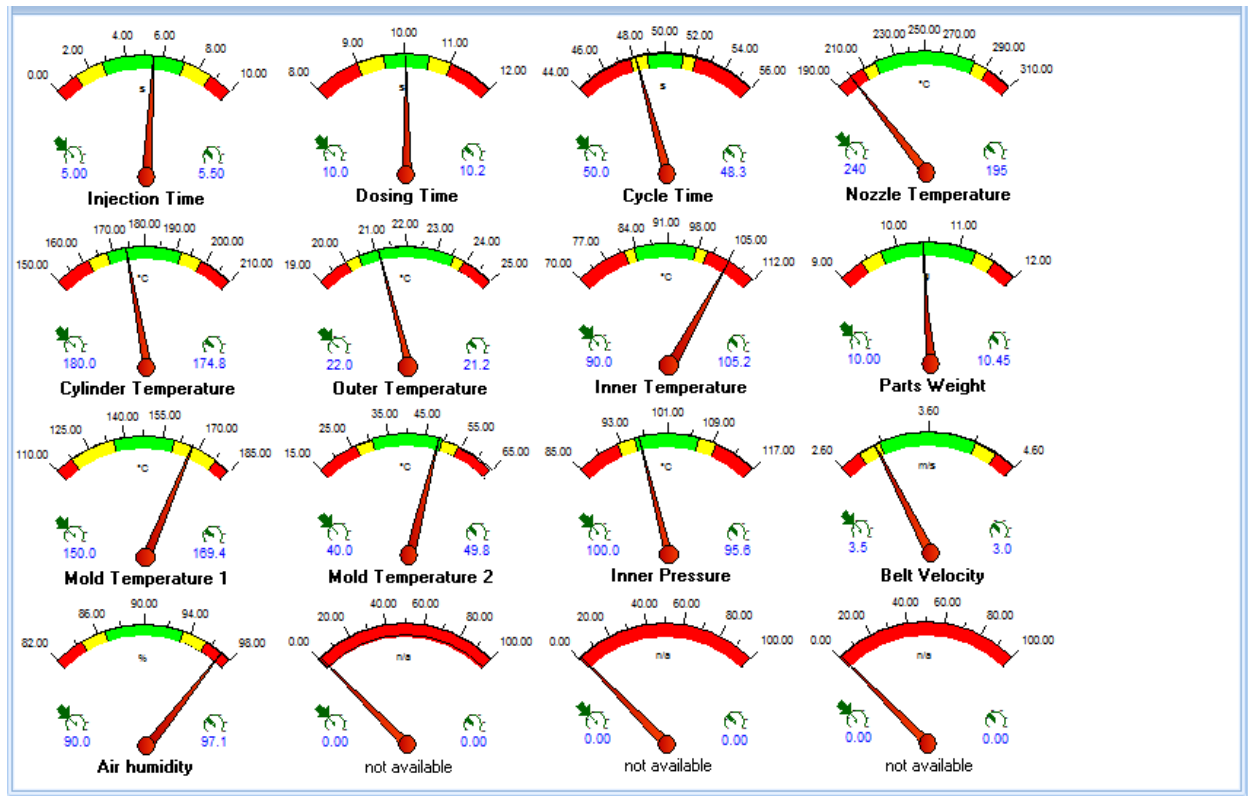
Process parameters

The process parameters drop down list includes a selection of process parameters.

Field descriptions

Detail applications

The selected process parameters pertaining to the selected workplace are displayed in the graphic of the visualization area.



5 Process Visualization

Summary

Menu	Quality management → Process analysis → Process visualization
Transaction code	vispd
Function authorization	visupd

Process visualization is used to monitor the process data in quality management.

Usage

The collection processes defined through the collection rules allow to visualize the defined characteristics online. The related data server provides the characteristics identified for visualization to the network. The visualization components existing in this application connect to these servers and will show the measured values online.

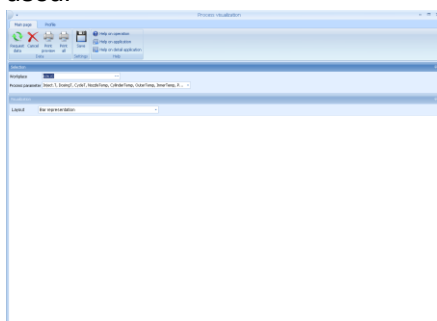
It is also possible to select the display layout and to integrate customized display layouts.

Integration

The display functions will visualize the data of the involved shop floor servers in accordance with the collection specifications.

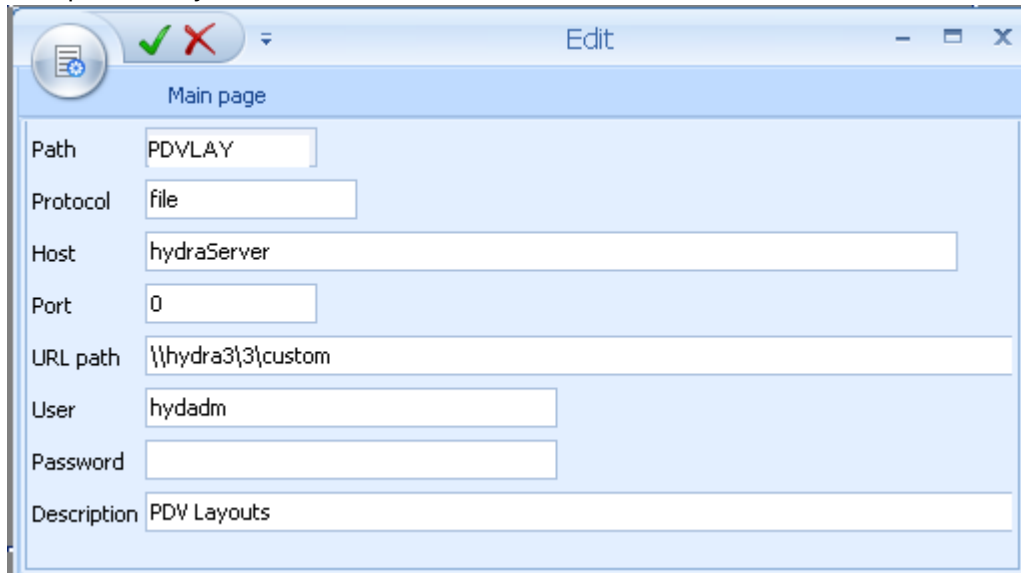
Requirements

It must be possible to reach the shop floor servers involved in the network and they must be configured for the presentation of the process data. A registration service (yellow pages to visualize process data online) must be set-up in the network. The display layouts must be locally available. If they are not saved locally they will be transferred from the directories of the corresponding HYDRA server to MOC upon the user's request. To do so, the button "Load PDV layouts from server" must be used.



In order to load the data from the server, the HYDRA path "PDVLAY" must be configured. HYDRA paths can be created using the MOC system under "System administration-> System settings -> Paths".

URL path = ./<system no.>/custom/



Selection criteria

The following selection criteria are available in the application:

Workplace

The pool application workplace can be used to select the desired workplace.

Process parameter

The process parameter drop-down list offers a selection of process parameters.

Field descriptions

Detail applications

The visualization area is used to show the graphs of the selected workplace and the individually selected process parameters. The display offers different visualization options that can be selected from the drop-down layout list. Several standard layouts are stored here but also special machine layouts created for the installations and machines.

Examples:

